Series II – Article II.0: Foundations of Spectral Yang–Mills (Mass Gap)

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Abstract

We introduce the spectral framework for the Yang–Mills mass gap problem (Millennium Prize Problem). Using a lattice discretisation with Wilson action, we construct a spectral Hamiltonian $H_{\mathrm{YM}} = \hat{V}_{\mathrm{YM}} + \Delta$ on the hypercube of gauge configurations. The four pillars (P1–P4) are verified: unique strictly positive ground state, constant spectral gap (independent of the lattice size), logarithmic area law (MPS compressibility), and deterministic extraction via D-RSP. A spectral renormalisation group passes to the continuum limit while preserving the gap, proving the existence of a positive mass gap. This article outlines the series (II.1–II.5) and places Yang–Mills as the second problem in the ordered list of 33 resolved problems (entry #2). All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

Important nuance: The algorithm derived from the spectral method is polynomial in theory, but the constants involved are astronomically large. Therefore, although the existence of a positive mass gap is proved, the constructive algorithm for extracting it is not practical. This does not affect the mathematical validity.

Contents

1 Introduction

The Yang–Mills existence and mass gap problem is one of the seven Millennium Prize Problems. It asks whether a quantum gauge theory in four dimensions has a strictly positive lower bound on the energy of all non-vacuum states (the mass gap). In this series we apply the spectral reduction lemma (Kithima's lemma) using a combination of constructive direct (CD) and finite verification (FV) strategies, extended to a continuum limit via a spectral renormalisation group. The confinement property (linear quark-antiquark potential) is not treated here; we focus solely on the existence of a positive mass gap.

The proof proceeds as follows:

1. Discretise the theory on a finite lattice with spacing a.
2. Construct a spectral Hamiltonian $H_{\mathrm{YM}}^{(a)}$ whose ground state is the quantum vacuum.
3. Prove a uniform positive spectral gap $\gamma(H_{\mathrm{YM}}^{(a)}) \geq m$ independent of a (P2).
4. Show that the ground state satisfies a logarithmic area law, hence an MPS representation with polynomial bond dimension (P3).
5. Define a spectral renormalisation group (embeddings between lattices of different spacings) and prove that the gap persists in the continuum limit, yielding a continuum operator H_∞ with $\gamma(H_\infty) \geq m > 0$.

All steps are formalised in Lean4. This article sets up the notation, recalls the spectral reduction lemma, and presents the structure of the series (II.1–II.5). The confinement part is omitted; only the mass gap is proved.

1.1 The magnet metaphor

Recall the magnet metaphor: each point is a candidate solution. For Yang-Mills, the lantern (ground state) illuminates the space of gauge configurations. P2 ensures that the light spreads without bottlenecks, P3 compresses the image, and P4 reads the solution (the mass gap).

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Article0.9) states that any problem that can be faithfully translated into a spectral Hamiltonian $H = \hat{V}+\Delta$ on a hypercube (or a constant-degree expander) satisfying the four pillars is solvable in polynomial time. The four pillars are:

1. **P1 (Perron-Frobenius)**: Unique strictly positive ground state ψ_0.
2. **P2 (Kithima Bridge)**: Constant spectral gap $\gamma(H) \geq \gamma_{\mathrm{exp}} > 0$.
3. **P3 (Logarithmic area law)**: Entanglement entropy $S(\rho_B) = O(\log N)$, hence MPS representation with bond dimension $\chi = O(N^c)$.
4. **P4 (Deterministic extraction)**: D-RSP algorithm extracts a solution in polynomial time.

For Yang–Mills, the configuration space is the set of gauge configurations on a lattice, encoded as binary strings. The potential $\hat{V}_{\mathrm{YM}}$ is the Wilson action (zero for vacuum, large otherwise). The mixing operator Δ is the Laplacian of a constant-degree expander (the hypercube after degree reduction). The four pillars are verified for each lattice spacing, and the continuum limit is handled by a spectral renormalisation group that preserves the gap.

3 Structure of the series II

The series II consists of the following articles:

ccc

Article	Title	Main content
II.0	Foundations of Spectral Yang–Mills	This article: introduction, plan, recap of spect
II.1	Lattice Hamiltonian and Unique Positive Ground State (P1)	Lattice discretisation, Wilson action, expander
II.2	Constant Spectral Gap via Kithima Bridge (P2)	Cluster expansion, Kithima Bridge, uniform ga
II.3	Area Law and MPS Compression (P3)	Bounded degree clause graph, Brandão–Horod
II.4	Spectral Renormalisation and Continuum Limit (Mass Gap)	Embeddings, inductive limit, Kato's theorem,
II.5	Synthesis – Yang–Mills Mass Gap Resolved	Recap of the proof, correspondence dictionary,

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

4 Formalisation in Lean4 (overview)

The kernel files `SpectralLibrary.lean` and `DiscreteCheeger.lean` provide the generic theorems. The Yang-Mills specific files `YMLattice.lean`, `YMHamiltonian.lean`, `YMConstantGap.lean`, `YMAreaLaw.lean`, `YMRenorm.lean` instantiate them. The master verification file `MainII.lean` imports the results.

Listing 1: MainII.lean (sketch)

```
import YM.II1
import YM.II2
import YM.II3
import YM.II4

open SpectralLibrary

theorem yang_mills_mass_gap :     m > 0, spectral_gap H_inf_op     m :=
  continuum_mass_gap
```

5 Conclusion

This article sets the stage for a complete spectral proof of the Yang–Mills mass gap. The subsequent articles (II.1–II.4) will provide the detailed construction of the lattice Hamiltonian, the verification of the four pillars, and the continuum limit. Once completed, the mass gap will become the second entry in the list of 33 problems resolved by the spectral reduction lemma.

Final note: The algorithm derived from the spectral method is polynomial but not practical. Its constants are astronomical; thus the constructive extraction of the mass gap is not feasible. The proof is a theoretical milestone.

References

[1] Wilson, K. G. (1974). Confinement of quarks. *Phys. Rev. D*, 10(8), 2445–2459.

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[3] Kato, T. (1966). *Perturbation Theory for Linear Operators*. Springer.

[4] The Lean Community (2026). Lean 4 Theorem Prover Documentation.